Gold Miner

A Fair Mining Spatial Competition Game on X1

On-chain multiplayer grid mining with
consumable gold spots and protocol-owned liquidity.

Litepaper — Draft September 10, 2026

Abstract

Gold Miner is an on-chain multiplayer fair mine game deployed on the X1 Network (Solana VM). Players navigate a 1,024 × 1,024 grid, discovering and mining gold spots that yield GOLD tokens (Token-2022). Gold spots are consumable — once mined, they never respawn for that world epoch. This creates a spatial competition where players must balance exploration, path optimization, and speed against other participants. All minted GOLD that is routed through the treasury goes to protocol-owned liquidity and is permanently burned, creating a deflationary token economy.

Key properties: permissionless entry, deterministic world generation, session-key gameplay, auto-LP burn treasury, and permissionless bitmap reset at 75% exhaustion.

Contents

1 Introduction

Most blockchain games fall into two categories: fully on-chain but mechanically simple (turn-based, deterministic), or visually rich but mostly off-chain (centralized state, token-gated). Gold Miner pursues a third path: a real-time spatial competition game where every move is recorded on-chain, yet the player experience is fluid and responsive.

The game is played on a 1,024 × 1,024 grid. Each cell is either empty or contains gold, determined by a public formula: $(x \ \& \ y) \ \% \ 7 = 0$. This deterministic generation means every player sees the same world, and foresight — knowing where gold lies ahead — becomes a strategic tool.

Players begin at position $(1, 1)$. From there, they move using arrow keys or WASD. Each move is an on-chain transaction signed by an ephemeral session key, eliminating the need for a wallet popup on every action. When a player steps on a gold spot, the contract automatically mints 100 GOLD tokens (Token-2022) to their wallet.

2 Core Mechanics

2.1 The Grid

The game world is a discrete grid with the following properties:

Property	Value
Dimensions	1,024 × 1,024 cells
Total cells	1,048,576
Gold formula	$(x \ \& \ y) \ \% \ 7 = 0$
Gold per spot	100 GOLD
Total supply	Uncapped; ∼150 million GOLD per world epoch (at 100% grid exhaustion)

The bitwise-AND modulo formula produces a scattered but deterministic pattern. Players can compute gold locations locally, enabling strategic path planning without revealing information to others.

2.2 Consumable Resources

Gold spots are consumed on first mine. Once a cell has been mined, it yields no further GOLD for the remainder of the world epoch. This is the central competitive mechanism.

The consequence is immediate: multiple players converging on the same region compete for a finite resource. Early arrival is rewarded not by a multiplier but by *access to unmined territory*. Late arrivals find only empty cells.

2.3 Spatial Diffusion Incentive

The Nash equilibrium of the game is for players to spread out across the map. Clustering is suboptimal because:

- Shared finite spots lower per-player yield
- Travel time to unmined areas increases with local density
- First-mover advantage in any region belongs to whoever arrived first

This mirrors real-world resource competition — gold rushes, mining claims, territorial exploration — where knowledge and speed compound.

2.4 Session Keys

To avoid requiring a wallet signature on every move, Gold Miner uses ephemeral session keys:

1. Player connects wallet and signs one transaction to join
2. Contract generates an encrypted session keypair (AES-256-GCM via Web Crypto API)
3. Session key signs all subsequent move transactions
4. Session expires after ∼4 hours (36,000 slots)
5. Player can start a new session at any time with one wallet signature

This preserves non-custodial security while enabling fluid gameplay.

3 Token Economics

3.1 GOLD Token

GOLD is a Token-2022 (SPL Token-2022 program) mint with the following parameters:

Parameter	Value
Name	Goldium
Symbol	GOLD
Decimals	9
Mint authority	Game program (mint-on-demand)
Freeze authority	None

Tokens are minted exclusively through gameplay. There is no pre-mine, no airdrop, no founder allocation. The only source of new GOLD is mining gold spots on the grid.

3.2 Protocol Treasury and Auto-LP Burn

All mined GOLD flows through the following pipeline:

1. Player mines gold → 100 GOLD minted to their wallet
2. Simultaneously, 100 GOLD minted to the protocol treasury

3. Treasury automatically swaps GOLD for XNT via AMM
4. LP tokens are deposited into protocol-owned liquidity
5. LP tokens are permanently burned

This creates a deflationary flywheel: more gameplay → more treasury volume → deeper liquidity → lower slippage → better player experience.

3.3 Bitmap Reset

When 75% of gold spots have been mined, any player can call `reset_mineable_area()` to:

- Zero the on-chain bitmap tracking mined cells
- Reset the exhaustion counter
- Allow gold to respawn across the grid

This is permissionless and trustless. No admin key, no governance vote. The game self-resets when the world is sufficiently exhausted, preserving long-term playability.

4 Architecture

4.1 Smart Contract

The game is built with Anchor on Solana VM and deployed on X1. Key program instructions:

Instruction	Description
`initialize_game`	One-time setup: grid size, gold formula parameters, token config
`join_game`	Player onboarding: creates player state, initializes session key
`move_and_mine`	Core gameplay: validates move, checks gold spot, mints if eligible
`deposit_xnt`	Funds player gas account for movement fees
`withdraw_xnt`	Returns unused gas to player wallet
`reset_mineable_area`	Permissionless bitmap reset at 75% exhaustion
`treasury_auto_lp`	Converts treasury GOLD to protocol-owned LP + burn

4.2 On-Chain State

4.3 Frontend

The client is a Next.js application with TypeScript and TailwindCSS. It handles:

Account	Purpose
Game Config	Grid dimensions, gold formula, mint addresses, fee parameters
Player State	Position, mining history, session key metadata
Bitmap Account	Bitmask of mined cells (sparse, compressed)
Treasury ATA	Holds accumulated 10% fees before auto-LP conversion

- Wallet connection (Backpack, Phantom, Solflare)
- Session key generation and encryption
- Grid rendering with WebGL-accelerated canvas
- Move batching and RPC optimization
- Real-time position and balance updates

5 Security Model

5.1 Player Protections

- **Non-custodial:** Player retains full control of wallet and funds at all times
- **Session key expiry:** Ephemeral keys automatically expire, limiting scope of compromise
- **Encrypted storage:** Session keys are AES-256-GCM encrypted in browser localStorage
- **No admin drain:** No privileged instruction can transfer player funds

5.2 Game Integrity

- **Deterministic world:** Gold locations are computed on-chain from public formula — no oracle, no randomness, no manipulation
- **Consumable verification:** Bitmap account is updated atomically with mint — double-mining is impossible
- **Permissionless reset:** No single party controls world state transitions

5.3 Economic Safeguards

- **Mint-on-demand only:** No pre-mine, no hidden inflation
- **Protocol-owned liquidity:** Treasury funds are LP'd and burned, not held
- **Immutable config option:** Game parameters can be locked via `finalize_game()` to prevent post-launch changes

6 Gas Economics

Movement and mining incur on-chain compute costs. Based on actual testnet transactions:

Action	Fee (XNT)	Compute Units
Move (empty cell)	∼0.0012	∼33,500
Mine gold (Token-2022 mint)	∼0.0015–0.0017	∼53,000–57,000

Players pre-fund a gas account via `deposit_xnt`. Unused gas can be withdrawn at any time. At typical prices, 1 XNT funds approximately 600–800 moves.

7 Roadmap

Phase	Status	Description
Testnet Alpha	**Live**	Core gameplay, session keys, basic grid rendering
Testnet Beta	In Progress	Treasury auto-LP, bitmap reset, stress testing
Security Audit	In Progress	External review of contract and economic model
Mainnet Launch	Planned	Fresh Token-2022 mint, new program keypair, initial LP seed
Post-Launch	Planned	Leaderboards, seasonal resets, GOLD NFT utility

8 Conclusion

Gold Miner demonstrates that on-chain games need not sacrifice mechanical depth for trustlessness. By treating the blockchain as a deterministic, shared game server rather than a settlement layer, we achieve:

- True ownership of mined assets
- Verifiable scarcity and fair distribution
- Censorship-resistant gameplay
- Protocol-aligned incentives through auto-LP burn

The consumable-resource model ensures competition emerges endogenously from the rules rather than from artificial leaderboards or time-limited events. Every player starts at the same position with the same information. What separates them is speed, strategy, and execution.

Disclaimer

Not Financial Advice. This litepaper is for informational purposes only and does not constitute financial, investment, legal, or tax advice. Nothing contained herein should be construed as a recommendation to buy, sell, or hold any digital asset, security, or financial instrument. Readers are strongly encouraged to conduct their own independent research ("DYOR") and consult with qualified professional advisors before making any financial decisions.

No Offer of Securities. The GOLD token and the Gold Miner game are experimental blockchain-based applications. This document does not constitute an offer to sell, a solicitation of an offer to buy, or a recommendation of any security or investment product. Participation in the protocol is entirely voluntary and at the participant's sole risk.

Risk Factors. Blockchain-based games and tokens involve significant risks, including but not limited to: smart contract vulnerabilities, regulatory uncertainty, market volatility, liquidity risk, and potential total loss of funds. The authors, developers, and affiliated parties assume no liability for any losses incurred as a result of participating in the Gold Miner protocol.

Forward-Looking Statements. Certain statements in this litepaper may be forward-looking in nature, including references to the roadmap, future features, and planned upgrades. Such statements are based on current expectations and assumptions that are subject to risks and uncertainties. Actual results may differ materially from those expressed or implied. The development team reserves the right to modify, delay, or cancel any planned feature at its sole discretion.

Experimental Technology. Gold Miner is built on emerging blockchain infrastructure (X1 Network / Solana VM). The protocol has not been formally audited at the time of publication. Users should treat all interactions with the protocol as experimental and exercise extreme caution.

Third-Party Dependencies. The protocol relies on external services, wallet providers, RPC endpoints, and oracle mechanisms that are outside the control of the development team. No guarantee is made regarding the availability, accuracy, or security of these third-party services.

No Warranties. THE PROTOCOL, THIS LITEPAPER, AND ALL ASSOCIATED MATERIALS ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, OR NON-INFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES, OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT, OR OTHERWISE, ARISING FROM, OUT OF, OR IN CONNECTION WITH THE PROTOCOL OR THE USE OR OTHER DEALINGS IN THE PROTOCOL.

This litepaper is a living document. For the latest version, contract addresses, and frontend access, see the official repository or join the community channels.